

Lecture 4

EE-215 Electronic Devices and Circuits

Asst Prof Muhammad Anis Ch

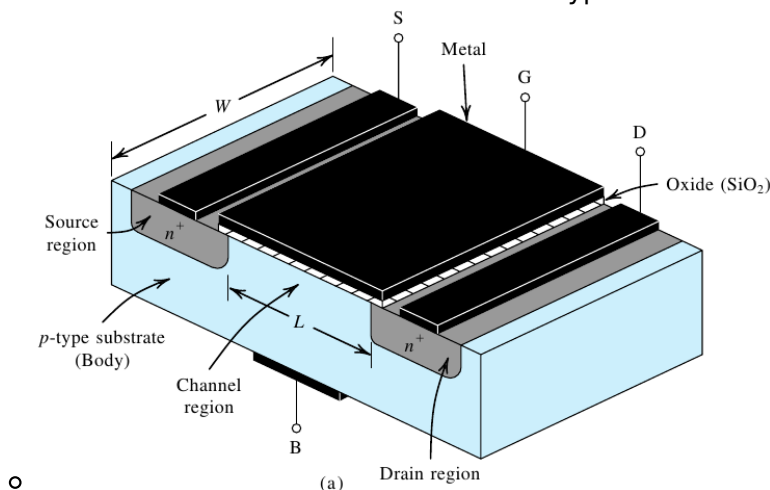
Three-Terminal Semiconductor Devices

- Three-Terminal Semiconductor Devices
 - Metal-oxide-semiconductor field-effect transistor (MOSFET)
 - Bipolar Junction Transistor (BJT)
- In case of a three terminal device,
 - a voltage applied between two terminals
 - can control the current flowing through the third terminal.
 - thus a three-terminal device can be used to realize a controlled current source (dependent source),
 - which is the basis for amplifier design
 - and a switch design (i.e. the control signal can be used to change the current in the 3rd terminal from zero to a large value) $\Rightarrow$ the 3-terminal device can be used as a switch

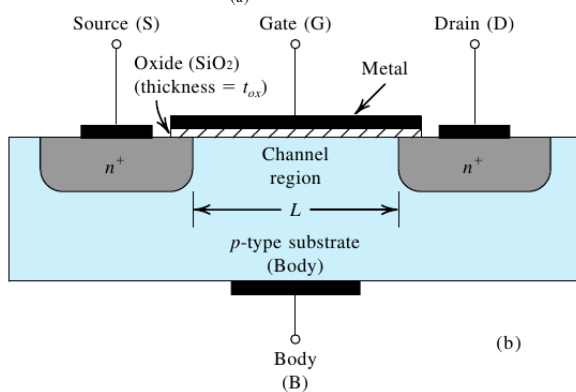
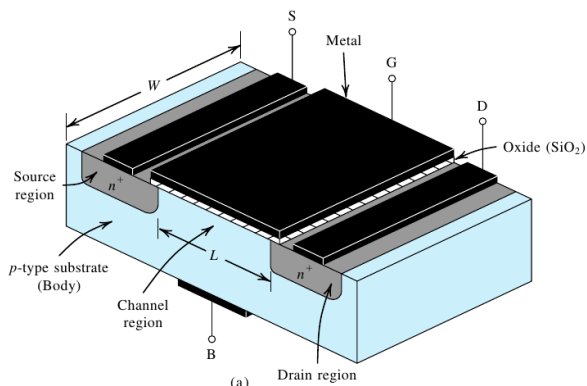
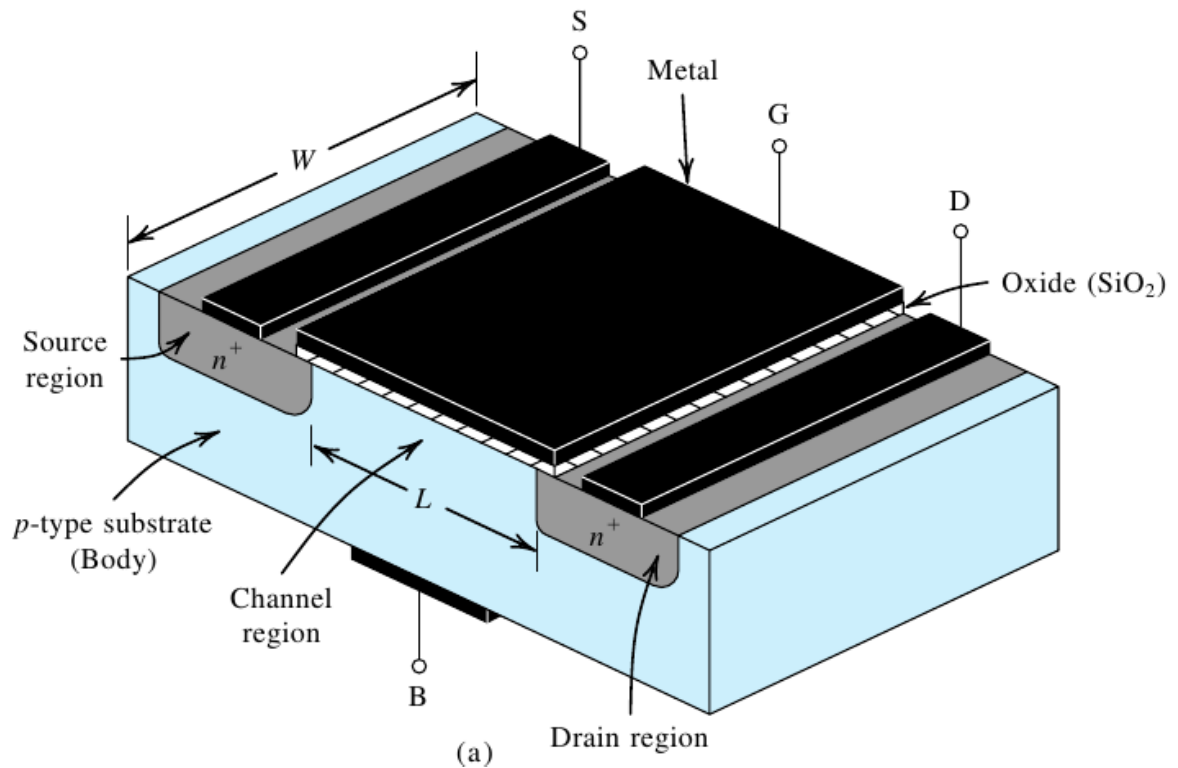
MOS Field Effect Transistor (MOSFETs)

Device Structure and Physical Operation

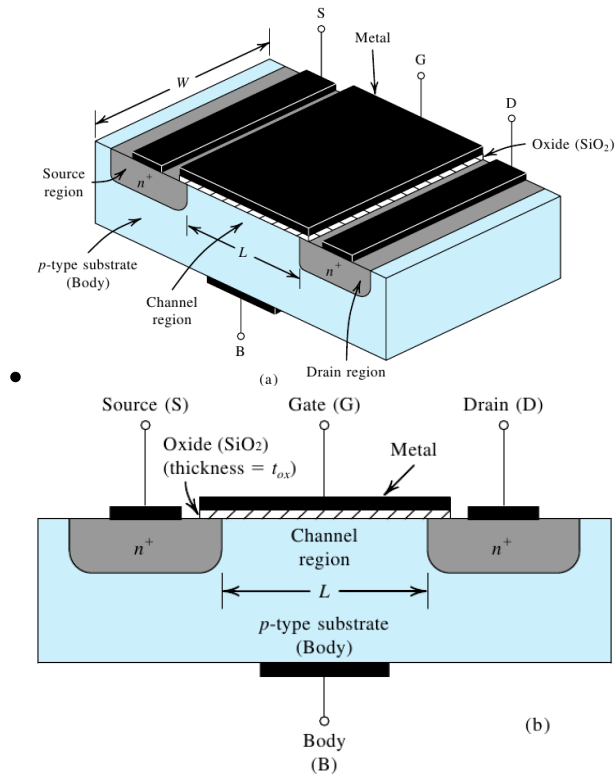
- The enhancement-type MOSFET is the most widely used field effect transistor.
 - it can be
 - an n-channel enhancement type MOSFET
 - or a p-channel enhancement type MOSFET
 - let's start with n-channel enhancement type MOSFET



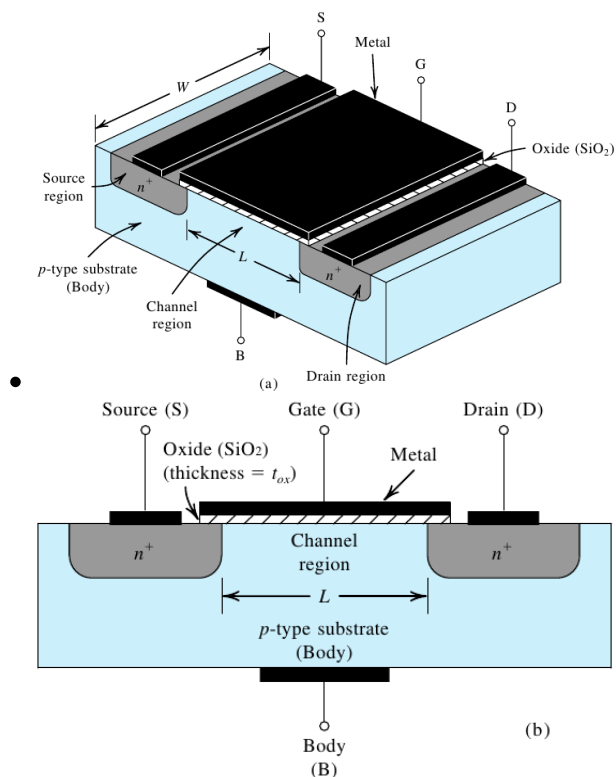
Device Structure



- this transistor is fabricated on a *p*-type substrate
- two heavily doped *n*-type (n^+) regions are created in the substrate
 - called the source (S) and the drain (D)
- A thin layer of Silicon dioxide (SiO_2) of thickness t_{ox} (typically 1 to 10nm)
 - is grown on the surface of the substrate.

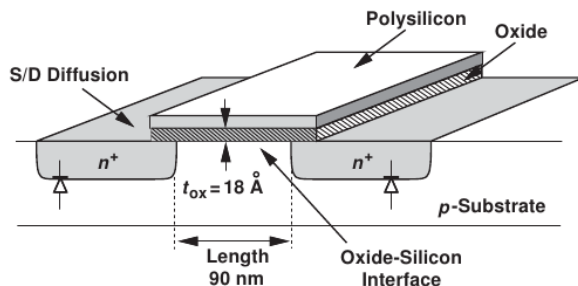


- SiO_2 covers the area between the source and the drain regions and is an electrical insulator
- Metal is deposited on top of the oxide layer to form the gate electrode (G) of the device
- Metal contacts are also made to the source region, the drain region and the substrate
- the substrate is also called body

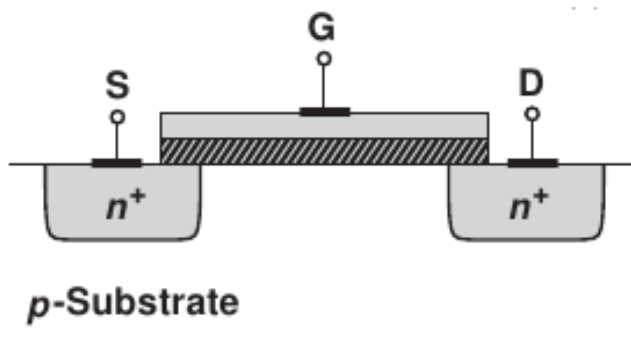


- thus this device has four terminals
 - the Gate Terminal, G

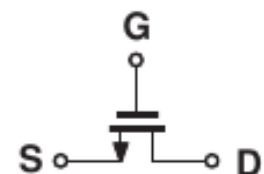
- the Source Terminal, S
- the Drain Terminal, D
- and the substrate or body terminal, B
- thus the name metal-oxide-semiconductor FET is derived from the physical structure of this device



- Note that the body forms pn junctions with the source and the drain regions
- in normal operation, these pn junctions are kept reverse-biased
 - by connecting body to the source terminal
 - Note that the drain terminal is always positive w.r.t. source terminal
- when the body is connected to the source terminal,
 - the MOSFET can be considered as a 3-terminal device
 - and its 3 terminals are Gate (G), Source (S) and Drain (D)



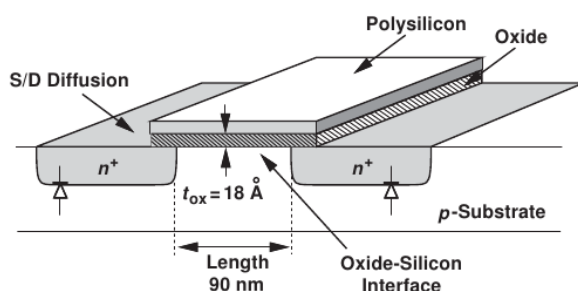
(b)



(c)

- the circuit symbol for an NMOS transistor is shown,
- wherein the arrow indicates the source terminal

Operation with zero Gate Voltage



- with zero voltage applied to the gate,
 - we have two reverse-biased back-to-back diodes,
 - that prevent current conduction from drain to source
- One diode is formed by the pn junction between p-type substrate and the n^+ drain region
- and the other diode is between the p-type substrate and the n^+ source region.

Creating a Channel for Current Flow

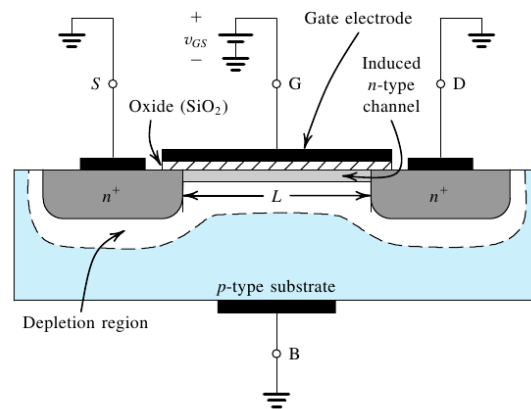
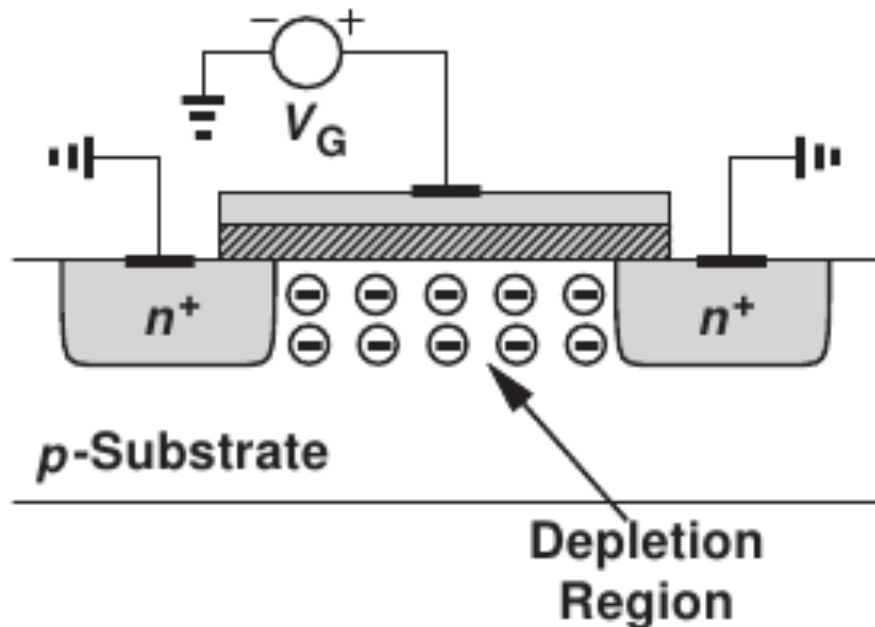


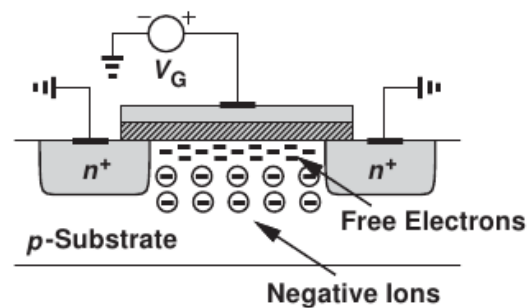
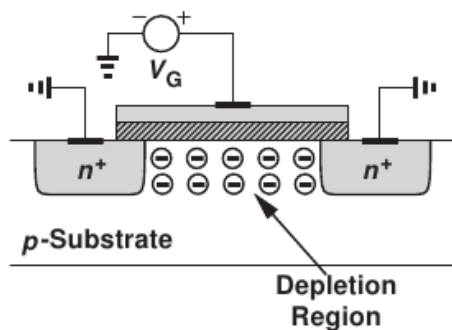
Figure 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

- consider the arrangement where source and drain are grounded
 - and the gate voltage is varied
- as the source is grounded this gate voltage appears
 - between gate and source and is denoted as v_{GS}
- As gate voltage v_{GS} increases from zero,
 - the positive charge on the gate repels the holes in the substrate,
 - thereby exposing negative ions and creating a carrier-depletion region



(b) formation of depletion region,

- Note that the device is acting as a capacitor
 - i.e. positive charge on the gate is mirrored by negative charge in the substrate
 - but no channel of mobile charges is created yet.
 - thus no current can flow from the drain to the source.
- thus the MOSFET is off



- (b) formation of depletion region,

- (c) formation of channel.

- As v_{GS} increases, the depletion region under the oxide becomes deeper
- when v_{GS} becomes sufficiently positive,
 - free electrons are attracted (from the n^+ source and the n^+ drain) to the oxide-silicon interface,
 - forming a conductive channel
- As a result, under the gate, an n region is created, connecting the drain and the source regions
- under the gate, an n region is created, connecting the drain and the source regions
 - now, if a voltage v_{DS} is applied, current will flow through this induced n region,
 - and the current is carried by the free-electrons in the induced region.

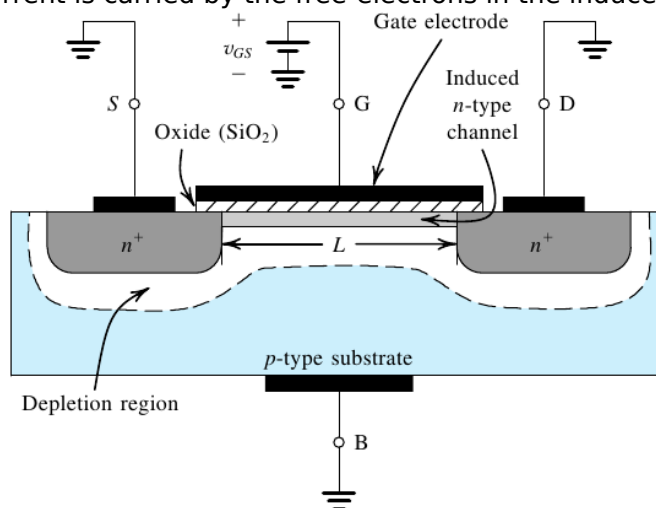


Figure 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

- this induced n region thus forms a channel for current flow from drain to source
 - note that electrons are moving from source to drain
- thus this MOSFET is called an n-channel MOSFET or an NMOS transistor

Threshold Voltage, V_t

- the value of v_{GS} at which a sufficient number of free electrons are present in the channel region,
 - to form a conducting channel is called the threshold voltage and is denoted by V_t
 - Note that V_t for an n-channel MOSFET is positive
 - and typically lies in the range of 0.3V to 1V
- Note that a parallel plate capacitor is formed by the gate
 - and the channel region of the MOSFET,

- with the oxide layer acting as the capacitor dielectric.
- The positive gate voltage causes positive charge to accumulate
 - on the top plate of the capacitor (the gate electrode),
 - the corresponding negative charge on the bottom plate is formed by the electrons in the induced channel
- thus an electric field exists in the vertical direction
 - this electric field controls the amount of charge in the channel,
 - and thus it determines the channel conductivity.
 - hence the name FET (Field Effect Transistor)

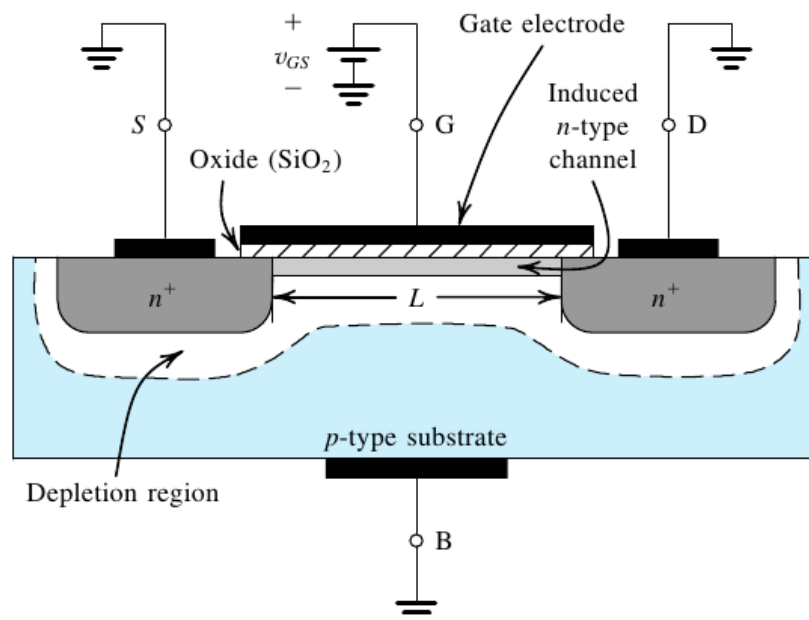


Figure 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

Overdrive Voltage, v_{OV}

- The voltage across the oxide (the parallel plate capacitor) must exceed V_t (the threshold voltage) to form a channel.
 - the excess of v_{GS} over V_t is called effective voltage or the overdrive voltage
 - i.e. $v_{OV} = v_{GS} - V_t$
 - the magnitude of the electron charge in the channel is given as
 - $|Q| = C_{ox}(WL)v_{OV} = C_{ox}(WL)(v_{GS} - V_t)$
 - because no mobile charge exists for $v_{GS} < V_t$
 - where, C_{ox} is called oxide capacitance, and is the capacitance per unit gate area (F/m^2)
 - W is the width of the channel and L is the length of the channel
- Oxide Capacitance, $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$
 - where
 - ϵ_{ox} is the permittivity of the silicon dioxide
 - $\epsilon_{ox} = 3.9\epsilon_o = 3.9 \times 8.854 \times 10^{-12} = 3.45 \times 10^{-11} F/m$
 - t_{ox} is the oxide thickness

Applying a Small, v_{DS}

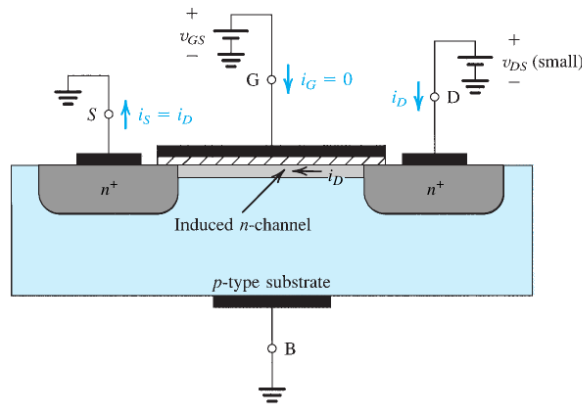


Figure 5.3 An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied. The device acts as a resistance whose value is determined by v_{GS} . Specifically, the channel conductance is proportional to $v_{GS} - V_t$, and thus i_D is proportional to $(v_{GS} - V_t)v_{DS}$. Note that the depletion region is not shown (for simplicity).

- now, a channel has been induced, by applying a +ve v_{GS}
- now if a +ve voltage v_{DS} is applied (assuming v_{DS} is small),
 - this voltage v_{DS} will cause a current i_D to flow through the induced n-channel

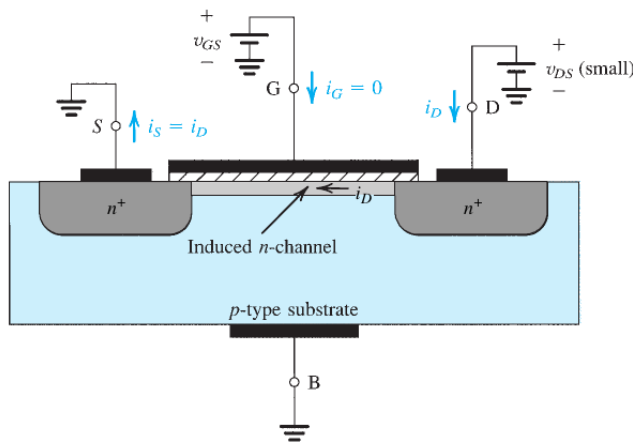


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- current is carried by free electrons travelling from source to drain (hence the names source and drain)
 - by convention, the direction for current, i_D is from drain to source
 - i.e. opposite to that of the flow of negative charges
- Now to calculate the value of i_D
 - As v_{DS} is small,
 - $\Rightarrow$ the voltage between the gate and various points along the channel remains almost constant
 - $\Rightarrow$ effective voltage between the gate and the various points along the channel remains equal to v_{OV}
 - $\Rightarrow |Q| = C_{ox}(WL)v_{OV}$
 - and $\frac{|Q|}{\text{unit channel length}} = C_{ox}Wv_{OV}$
 - The voltage v_{DS} establishes an electric field E across the length of the channel

$$(v = - \int E \cdot d\ell)$$
 - $|E| = \frac{v_{DS}}{L}$

- this electric field causes the channel electrons to drift towards the drain with a velocity of
 - *Electron drift velocity* $= \mu_n |E| = \mu_n \frac{v_{DS}}{L}$
 - where μ_n is the mobility of the electrons in the induced n-channel
 - i_D can be given as
 - $i_D = \frac{|Q|}{\text{unit channel length}} \times \text{Electron drift velocity}$
 - $i_D = (C_{ox} W v_{OV}) \left(\mu_n \frac{v_{DS}}{L} \right)$
 - or $i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \right] v_{DS}$
 - or $i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t) \right] v_{DS}$
 - $\Rightarrow$ for small v_{DS} , (for fixed $v_{OV} > 0$) the channel behaves as a linear resistance.
 - the value of this resistance is controlled by the overdrive voltage ($v_{OV} = v_{GS} - V_t$)
- $i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \right] v_{DS} = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t) \right] v_{DS}$
 - the conductance of the channel is
 - $g_{DS} = \frac{i_D}{v_{DS}} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t)$
 - where
 - $\mu_n C_{ox} = k_n'$ is determined by the process technology used to fabricate the MOSFET
 - and is called process transconductance parameter
 - and has units of A/V^2
 - $\frac{W}{L}$ is the transistor aspect ratio
 - W is the width of the channel
 - L is the length of the channel
- for a given fabrication process, there is a minimum channel length L_{min}
 - A fabrication process is often characterized by the L_{min} that is possible with that process
 - e.g. a 90nm process $\Rightarrow$ that for this process the minimum channel length possible is 90nm
 - MOSFET transconductance parameter, k_n is defined as
 - $k_n = k_n' \left(\frac{W}{L} \right) = (\mu_n C_{ox}) \left(\frac{W}{L} \right)$
 - and has units of A/V^2 as $\frac{W}{L}$ is a dimensionless quantity
- thus if v_{DS} is kept small, the MOSFET behaves as a linear resistance r_{DS}
 - whose value is controlled by the gate voltage v_{GS}
 - i.e. $r_{DS} = \frac{1}{g_{DS}} = \frac{1}{(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV}} = \frac{1}{(\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t)}$

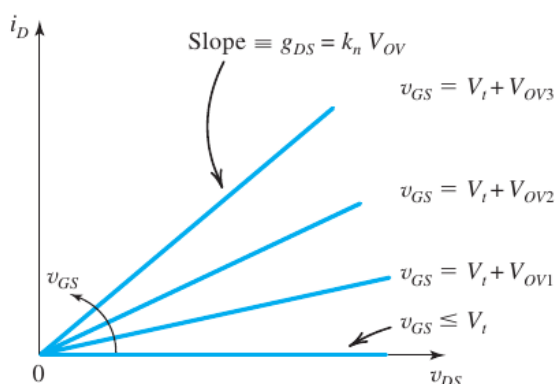


Figure 5.4 The i_D - v_{DS} characteristics of the MOSFET in Fig. 5.3 when the voltage applied between drain and source, v_{DS} , is kept small. The device operates as a linear resistance whose value is controlled by v_{GS} .

- So for a MOSFET to conduct, a channel has to be induced.
 - increasing v_{GS} above the threshold voltage V_t enhances the channel
 - therefore this transistor is called enhancement-type MOSFET

Operation as v_{DS} is Increased

- let v_{GS} be held constant at a value greater than V_t
 - i.e. MOSFET is operating at a constant overdrive voltage, v_{OV}

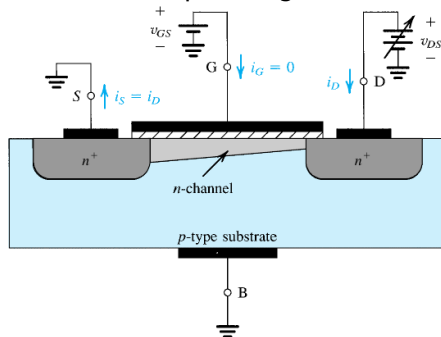
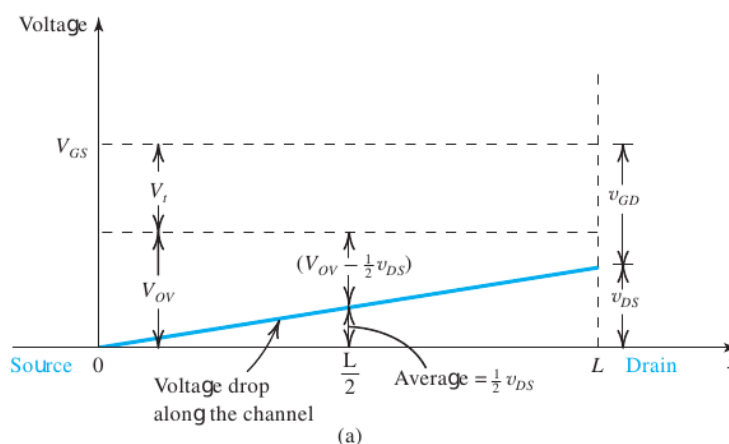


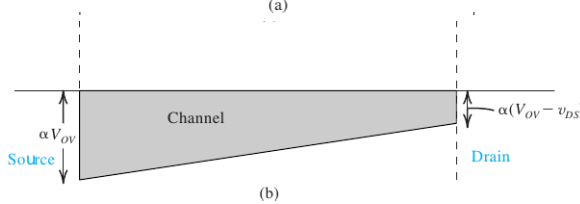
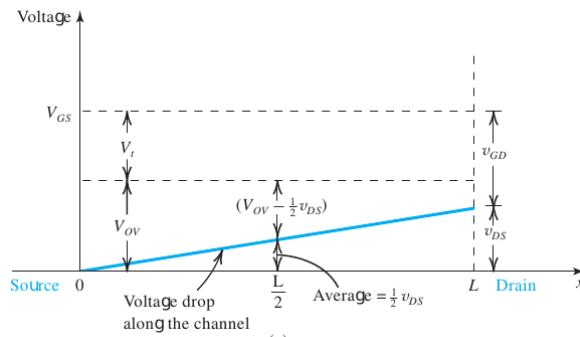
Figure 5.5 Operation of the enhancement NMOS transistor as v_{DS} is increased. The induced channel acquires a tapered shape, and its resistance increases as v_{DS} is increased. Here, v_{GS} is kept constant at a value $> V_t$; $v_{GS} = V_t + V_{OV}$.

- here v_{DS} appears as a voltage drop across the length of the channel.
- here v_{DS} appears as a voltage drop across the length of the channel.
 - i.e. as we travel along the channel from source to drain,
 - the voltage increases from zero to v_{DS}



- $\Rightarrow$ voltage between the gate and points along the channel decreases from $v_{GS} = V_t + v_{OV}$ at the source to $v_{GD} = v_{GS} - v_{DS} = V_t + v_{OV} - v_{DS}$ at the drain

- As the channel depth depends on how much this voltage exceeds V_t
 - $\Rightarrow$ excess voltage at source is v_{OV} while at load is $v_{OV} - v_{DS}$
 - thus the channel is not of uniform depth but is tapered shape



- channel is deepest at the source end ($\propto v_{OV}$) and shallowest at the drain ($\propto (v_{OV} - v_{DS})$)
- As v_{DS} is further increased, the channel becomes more tapered and its resistance increases
- As a consequence, the $i_D - v_{DS}$ curve is not a straight line but it bends (triode region)

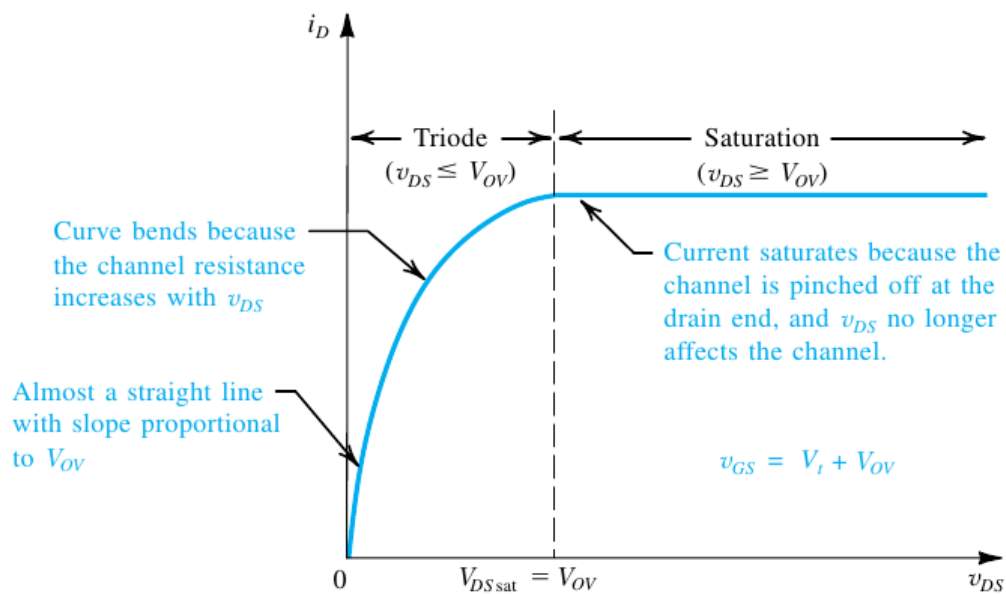
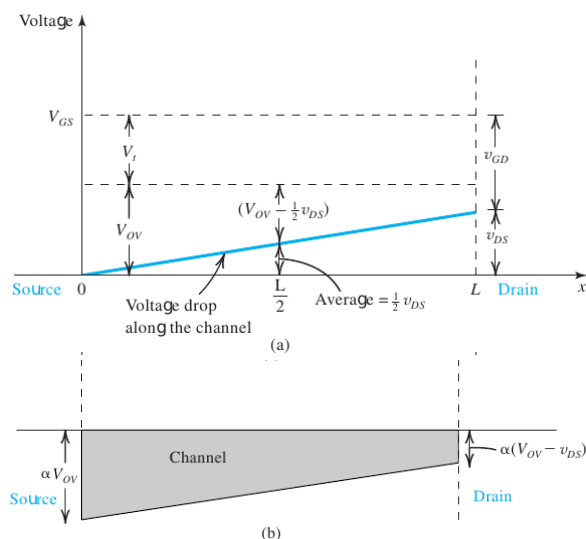


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} = V_t + V_{OV}$.



◦ As $i_D = \frac{|Q|}{\text{unit channel length}} \cdot \text{Electron Drift velocity}$

$$\circ i_D = \frac{|Q|}{\text{unit channel length}} \times \mu_n |E|$$

$$\circ i_D = \frac{|Q|}{\text{unit channel length}} \times \mu_n \frac{v_{DS}}{L}$$

◦ from fig

■ the average overdrive voltage is $v_{OV} - \frac{1}{2}v_{DS}$

$$\circ \Rightarrow i_D = C_{ox} W \left(v_{OV} - \frac{1}{2}v_{DS} \right) \mu_n \frac{v_{DS}}{L}$$

• or $i_D = \mu_n C_{ox} \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2}v_{DS} \right) v_{DS}$

$$\circ \text{ or } i_D = k_n' \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2}v_{DS} \right) v_{DS}$$

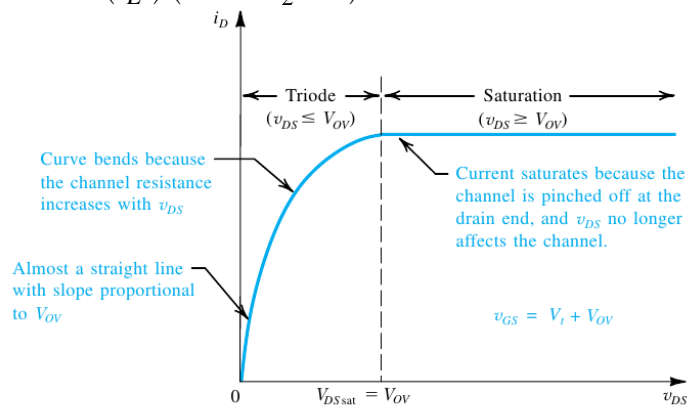


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS

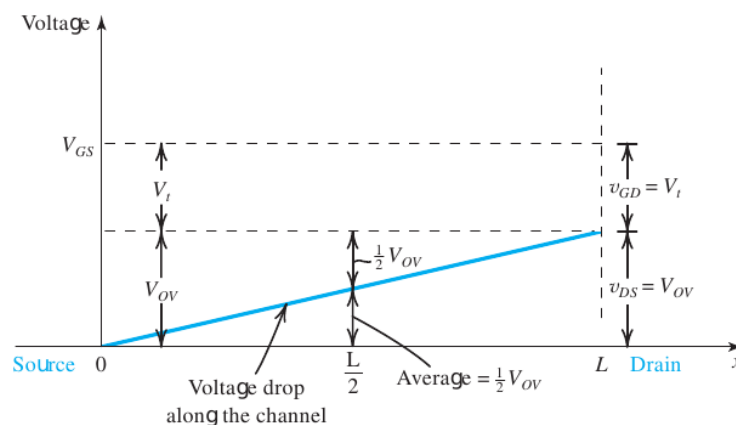
◦ transistor operated with $v_{GS} = V_t + V_{OV}$.

◦ Note that for v_{DS} very small $v_{OV} - \frac{1}{2}v_{DS} \approx v_{OV}$

$$\circ \Rightarrow g_{DS} = \frac{i_D}{v_{DS}} = k_n' \left(\frac{W}{L} \right) (v_{OV}) , \text{ for small } v_{DS} \text{ i.e. near the origin}$$

Operation for $v_{DS} \geq v_{OV}$

• for $v_{DS} = v_{OV}$, the channel depth at the drain is $v_{OV} - v_{DS} = 0$

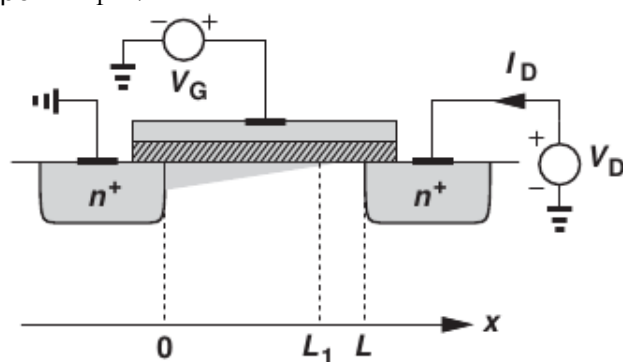


o (a)

- o the zero depth of the channel at the drain end gives rise to the term pinch off
- o thus the gate-substrate potential difference is not sufficient at $x=L$ to attract electrons and the channel is pinched off

- now when $v_{DS} > v_{OV}$

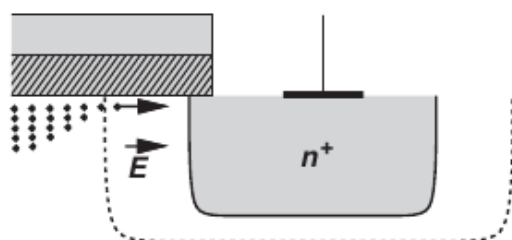
- o the voltage difference between the gate and the substrate falls to V_t at some point $L_1 < L$



o (b)

- o the device therefore contains no channel between L_1 and L

- the device still conducts



o (c)

- o once the electrons reach the end of the channel, they experience the high electric field
 - in the depletion region surrounding the drain junction and are rapidly swept to the drain terminal
- o however the drain voltage no longer affects the current significantly,
 - and thus the MOSFET acts as a constant current source
- thus when we increase v_{DS} beyond v_{OV} (i.e. for $v_{DS} > v_{OV}$),
 - o the current through the channel remains almost constant at
 - the value reached for $v_{DS} = v_{OV}$
 - o the drain current thus saturates at the value of

- $i_D = k_n' \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2} v_{DS} \right) v_{DS} \Big|_{v_{DS}=v_{OV}} = k_n' \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2} v_{OV} \right) v_{OV}$
- $i_D = k_n' \left(\frac{W}{L} \right) \left(\frac{1}{2} v_{OV} \right) v_{OV}$
- $i_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) v_{OV}^2$, (saturation region)
- and the MOSFET is said to have entered the saturation region of operation
- The voltage v_{DS} at which saturation occurs is
 - $v_{DSsat} = v_{OV} = v_{GS} - V_t$

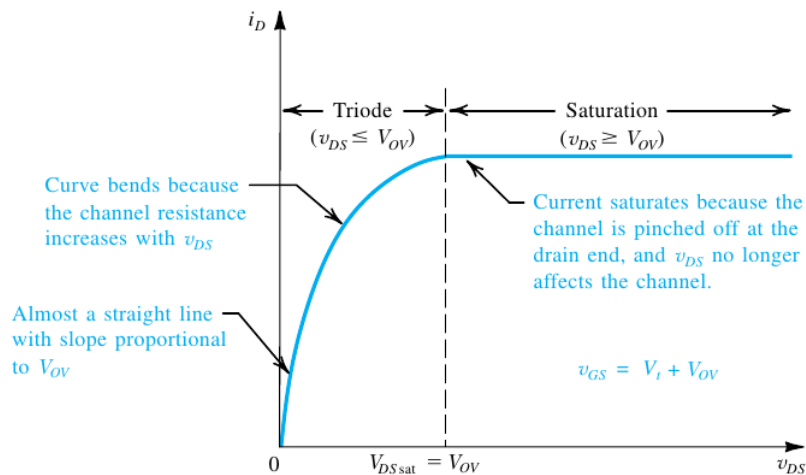
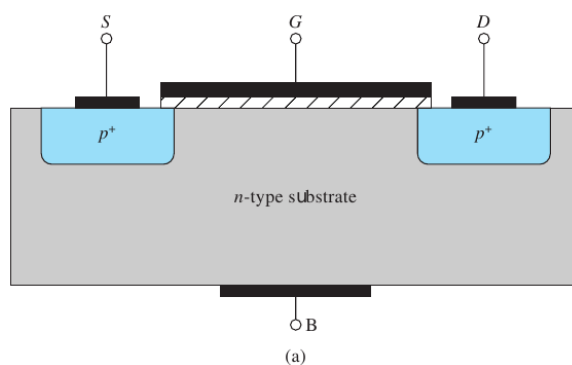
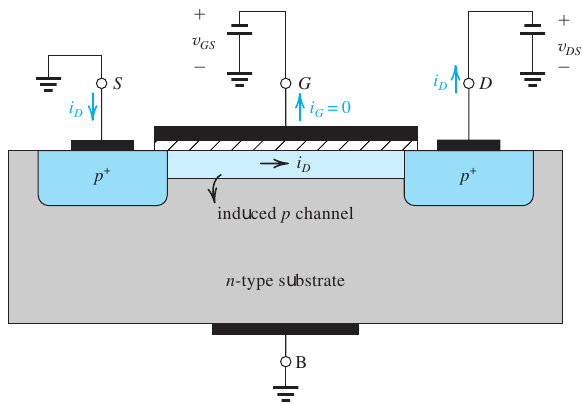


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS

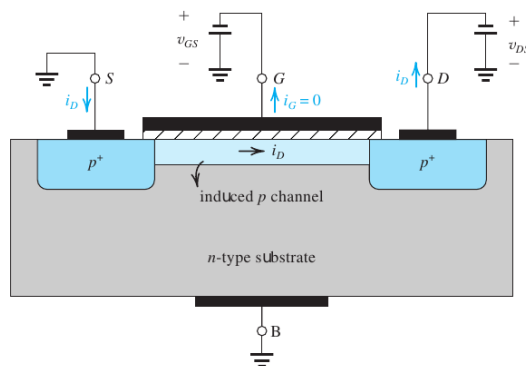
- transistor operated with $v_{GS} = V_t + V_{OV}$.
 - so the $i_D - v_{DS}$ curve is a horizontal straight line in the saturation region
 - the region of operation for non-pinched-off channel is called triode region
- thus for an n-channel MOSFET,
 - in the saturation region, $i_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) v_{OV}^2$
 - in the triode region, $i_D = k_n' \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2} v_{DS} \right) v_{DS}$

The p-Channel MOSFET





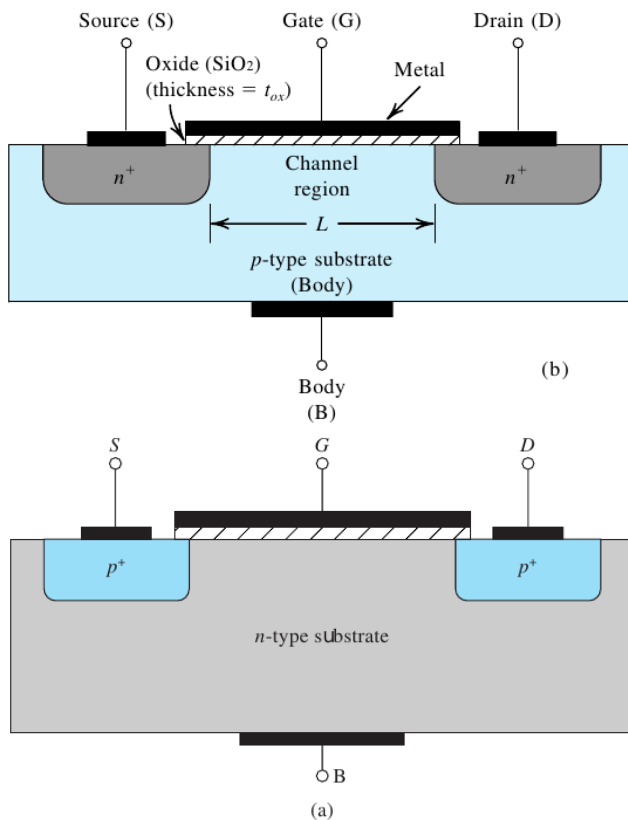
- here the substrate is n-type while the source and drain regions are p^+ type.
- i.e. all semiconductor regions are reversed in polarity relative to their counterparts in NMOS transistor
- the NMOS and PMOS transistors are called complementary devices
- for a PMOS device, the channel consists of holes and is formed
 - if the gate voltage is below the source potential by one threshold voltage
 - i.e. $v_{GS} \leq V_{tp}$ or $|v_{GS}| \geq |V_{tp}|$
 - to cause a current i_D to flow in the p-channel, a negative voltage v_{DS} is applied to the drain.
 - the current i_D is carried by holes and flows through the channel from source to drain



-
- the process transconductance for the PMOS devices is $k_p' = \mu_p C_{ox}$
 - μ_p is the mobility of the holes in the induced p-channel
 - typically $\mu_p = 0.25\mu_n$ to $0.5\mu_n$ and is process technology dependent
 - the transistor transconductance parameter is $k_p = k_p' \left(\frac{W}{L} \right)$

Complementary MOS or CMOS

Complementary MOS or CMOS Technology



- from figure,
 - the two types of MOSFETs require two different types of substrate
 - so is it possible to build both NMOS and PMOS devices on the same wafer?
 - fortunately, a local n-type substrate (called n-well) can be created in a p-type substrate, thereby accomodating PMOS transistors

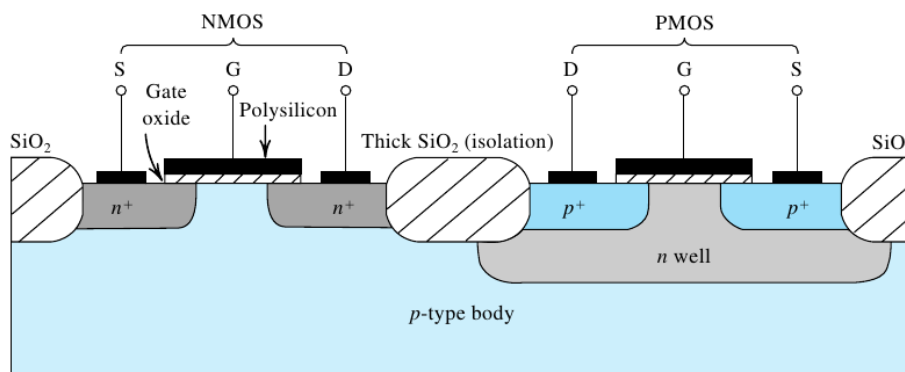
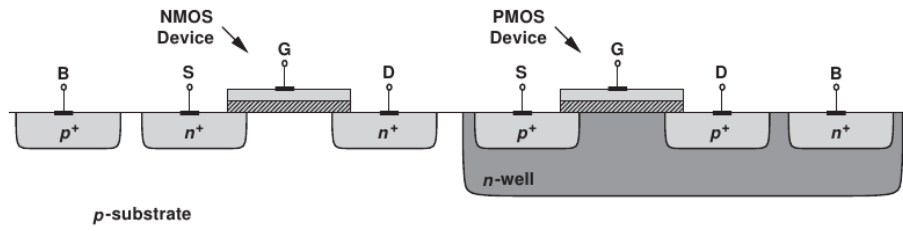


Figure 5.10 Cross-section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate n-type region, known as an n well. Another arrangement is also possible in which an n-type body is used and the n device is formed in a p well. Not shown are the connections made to the p-type body and to the n well; the latter functions as the body terminal for the p-channel device.

- - An n-well encloses a PMOS device while the NMOS transistor resides in the p-substrate



● Figure 6.35 CMOS technology.

- thus Complementary MOS technology employs MOS transistors of both polarities.
- At present CMOS is the most widely used technology.